

Humna Imran

Roll Number # 017

In [1]:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:

```
img_bgr = cv2.imread('/content/fruit.png')
```

In [3]:

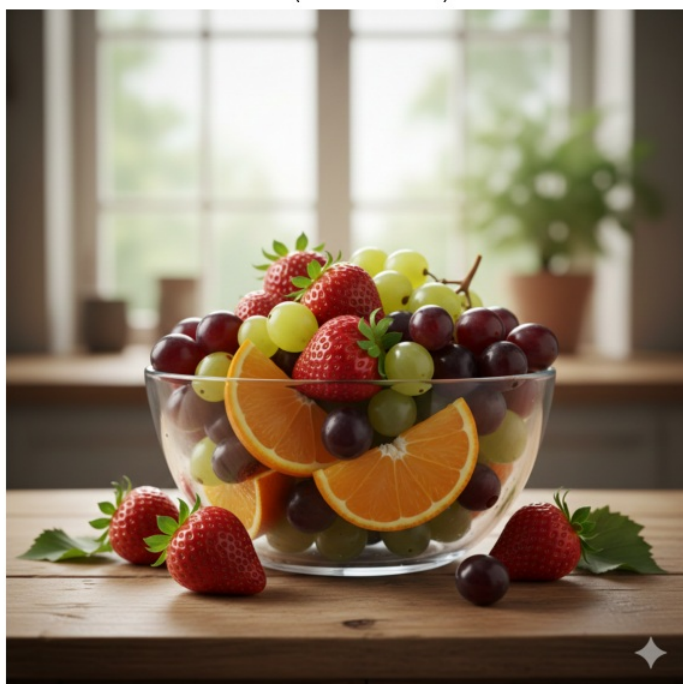
```
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)
```

In [4]:

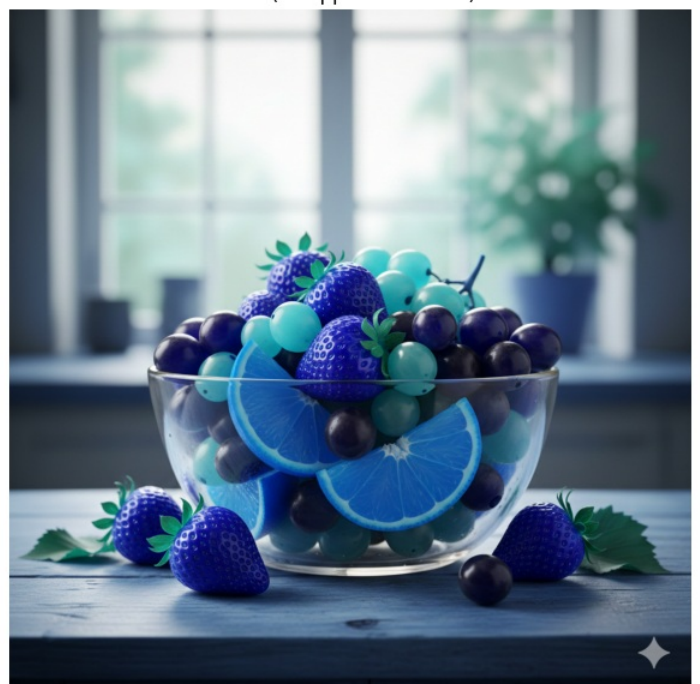
```
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title("RGB (Normal Colors)")
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(img_bgr)
plt.title("BGR (Swapped Channels)")
plt.axis('off')

plt.tight_layout()
plt.show()
```

RGB (Normal Colors)



BGR (Swapped Channels)



In [5]:

```
img_hsv = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2HSV)
img_lab = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2Lab)
```

```
img_ycrCb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2YCrCb)
```

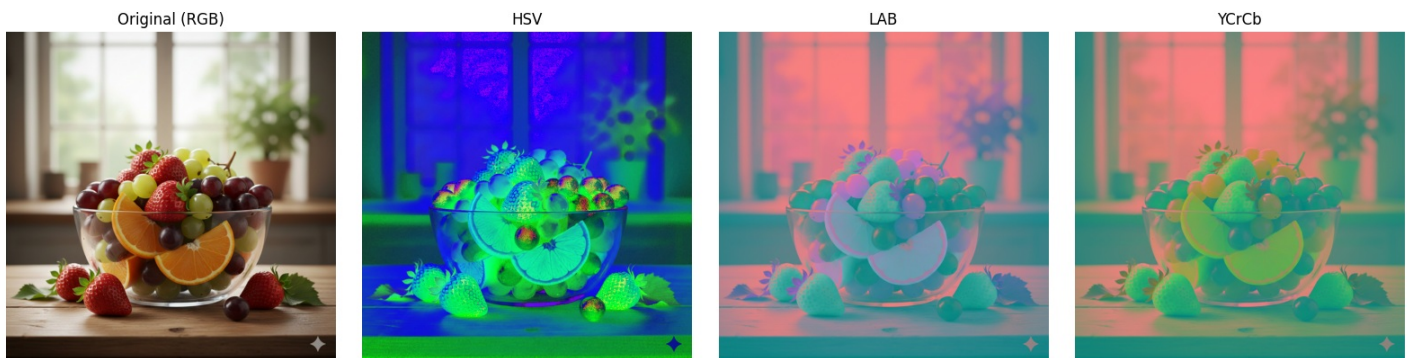
In [6]:

```
images = [img_rgb, img_hsv, img_lab, img_ycrCb]
titles = ["Original (RGB)", "HSV", "LAB", "YCrCb"]

plt.figure(figsize=(16, 4))

for i in range(4):
    plt.subplot(1, 4, i + 1)
    plt.imshow(images[i])
    plt.title(titles[i])
    plt.axis('off')

plt.tight_layout()
plt.show()
```



In [7]:

```
print(img_bgr.shape, img_rgb.shape)
print(img_bgr[100, 100], img_rgb[100, 100])

(1024, 1024, 3) (1024, 1024, 3)
[ 96 112 124] [124 112  96]
```

In [8]:

```
gray = cv2.cvtColor(img_bgr, cv2.COLOR_RGB2GRAY)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis('off')
```

Out[8]:

```
(np.float64(-0.5), np.float64(1023.5), np.float64(1023.5), np.float64(-0.5))
```

Grayscale Image





In [9]:

```
print(img_bgr.shape, img_rgb.shape)
img1 = img_bgr*[255,0,0]

(1024, 1024, 3) (1024, 1024, 3)
```

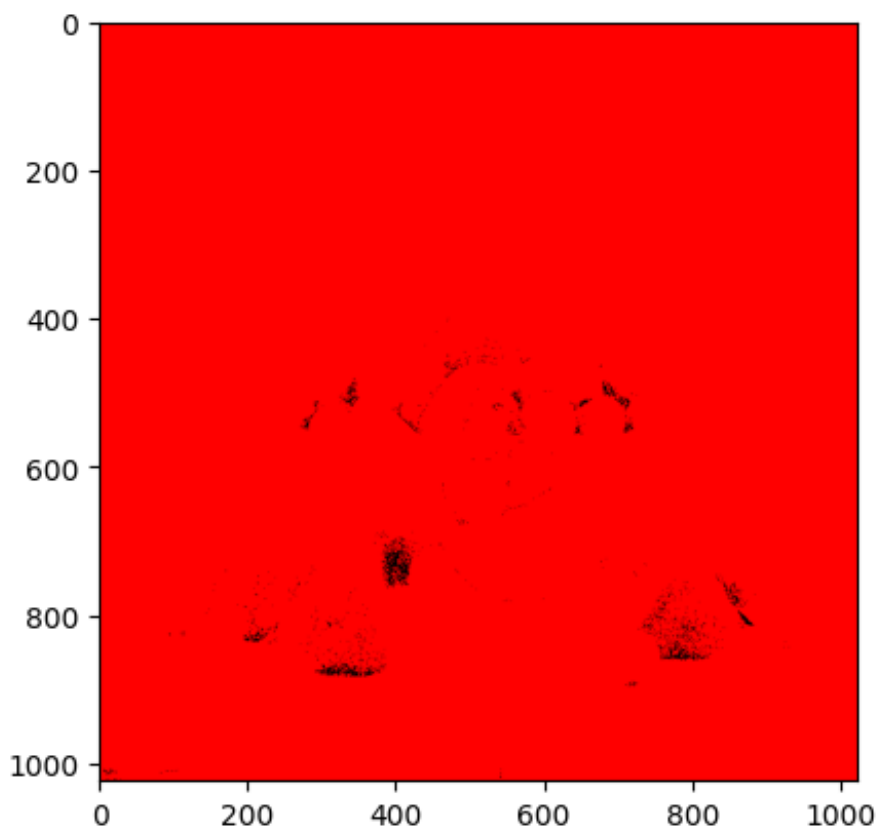
In [10]:

```
plt.imshow(img1)
```

WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers). Got range [0..65025].

Out[10]:

<matplotlib.image.AxesImage at 0x7d4259116180>



In [11]:

```
print(img_bgr.shape, img_rgb.shape)
img1 = img_bgr*[0,255,0]

(1024, 1024, 3) (1024, 1024, 3)
```

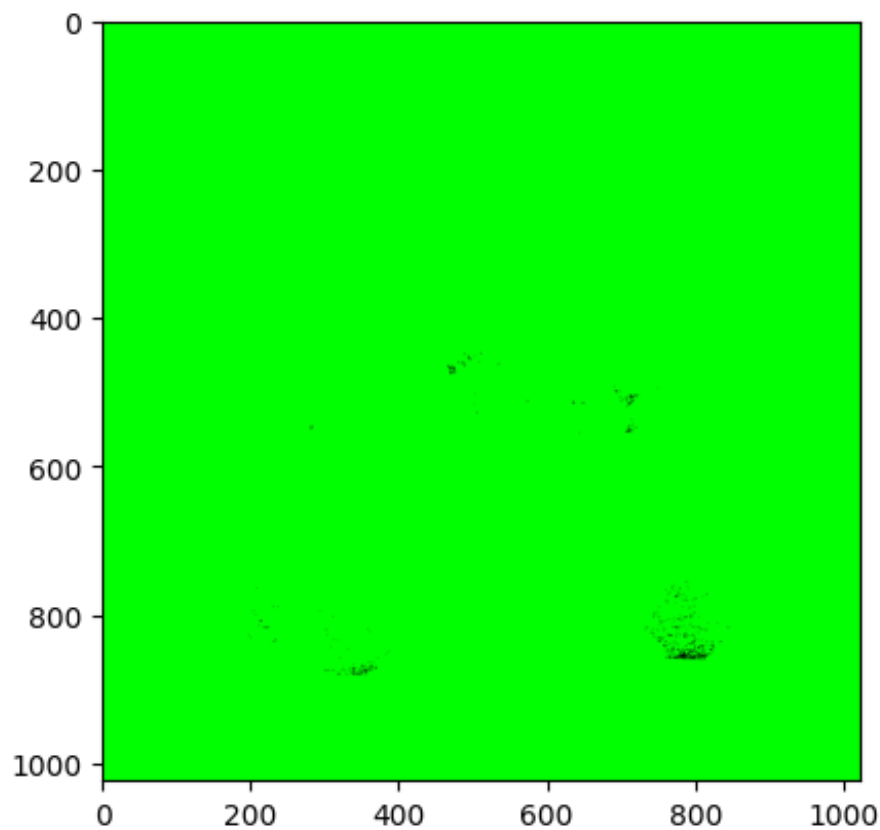
In [12]:

```
plt.imshow(img1)
```

WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers). Got range [0..65025].

Out[12]:

<matplotlib.image.AxesImage at 0x7d42591879b0>



In [13]:

```
print(img_bgr.shape, img_rgb.shape)
img1 = img_bgr*[0,0,255]
```

(1024, 1024, 3) (1024, 1024, 3)

In [14]:

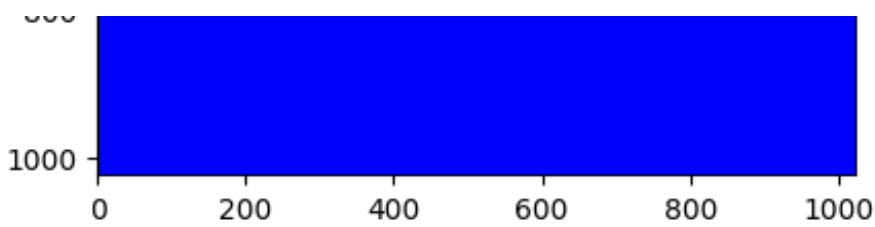
```
plt.imshow(img1)
```

WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers). Got range [0..65025].

Out[14]:

<matplotlib.image.AxesImage at 0x7d42591c9400>





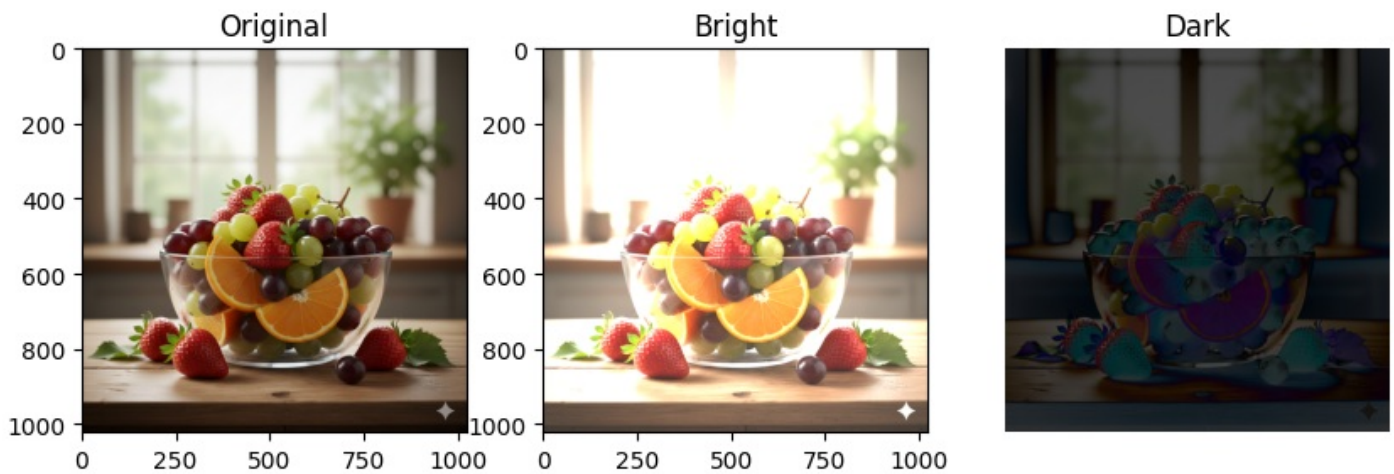
In [15]:

```
bright = cv2.convertScaleAbs(img_rgb, alpha=1.5, beta=50)
dark = cv2.convertScaleAbs(img_rgb, alpha=0.5, beta=-50)

plt.figure(figsize=(10,4))
plt.subplot(1,3,1); plt.imshow(img_rgb); plt.title("Original")
plt.subplot(1,3,2); plt.imshow(bright); plt.title("Bright")
plt.subplot(1,3,3); plt.imshow(dark); plt.title("Dark")
plt.axis('off')
```

Out[15]:

```
(np.float64(-0.5), np.float64(1023.5), np.float64(1023.5), np.float64(-0.5))
```



In [16]:

```
img_bgr = cv2.imread('/content/bgr.png')
```

In [17]:

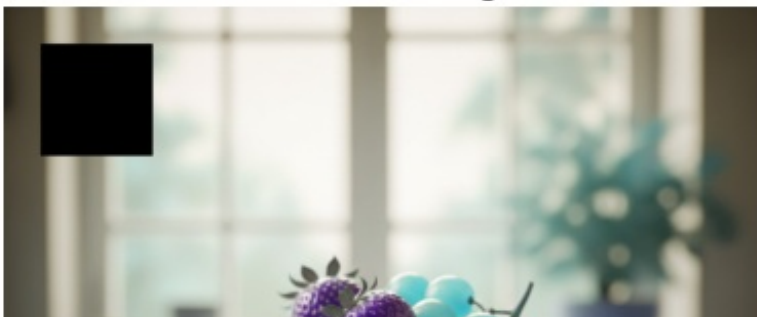
```
occluded = img_bgr.copy()
cv2.rectangle(occluded, (50,50), (200,200), (0,0,0), -1)

plt.imshow(occluded)
plt.title("Occluded Image")
plt.axis('off')
```

Out[17]:

```
(np.float64(-0.5), np.float64(1023.5), np.float64(1023.5), np.float64(-0.5))
```

Occluded Image





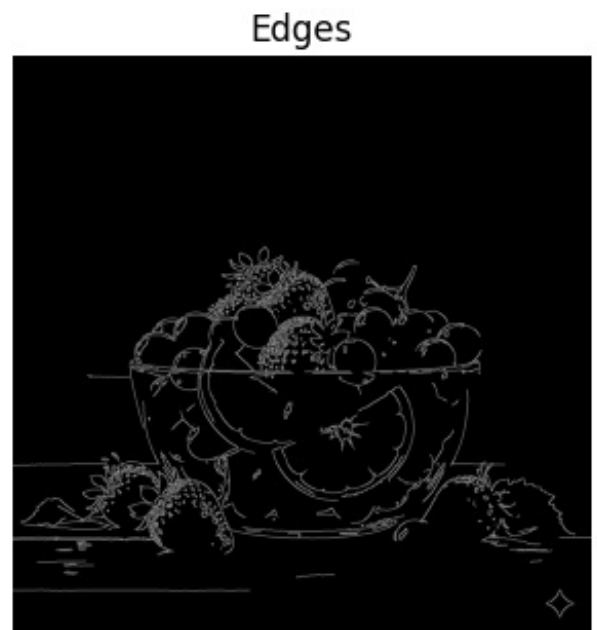
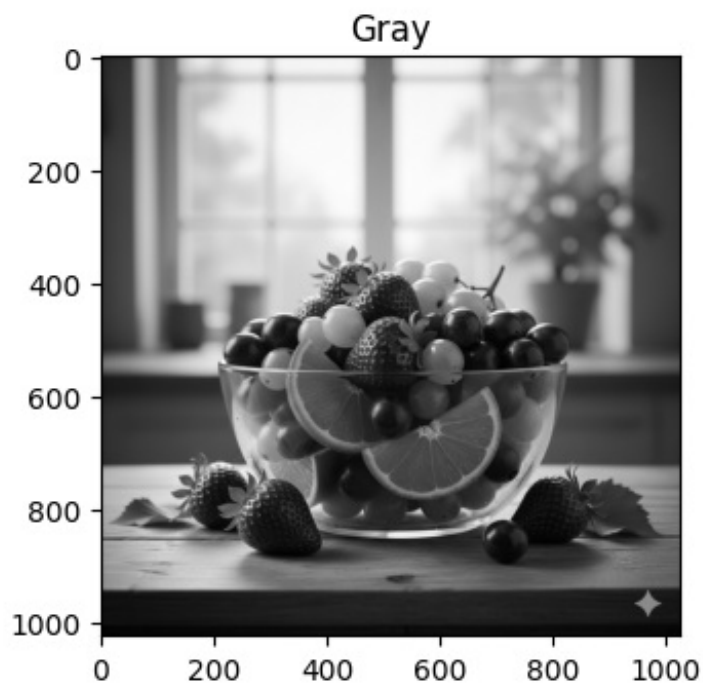
In [18]:

```
edges = cv2.Canny(gray, 100, 200)

plt.figure(figsize=(8,4))
plt.subplot(1,2,1); plt.imshow(gray, cmap='gray'); plt.title("Gray")
plt.subplot(1,2,2); plt.imshow(edges, cmap='gray'); plt.title("Edges")
plt.axis('off')
```

Out[18]:

```
(np.float64(-0.5), np.float64(1023.5), np.float64(1023.5), np.float64(-0.5))
```



In [19]:

```
cv2.imwrite("edges.jpg", edges)
```

Out[19]:

True